NASA's Lucy Spies a Wobbly, Peanut-Shaped Asteroid with a Watery Past

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When NASA's Lucy spacecraft zipped past the asteroid Donaldjohanson at 30,000 mph last year, it caught the space rock in a dizzying dance. Rather than spinning on a single axis like most planets and asteroids, this peanut-shaped body wobbles completing an end-over-end rotation every 10.5 days while swaying around its long axis every 26.5 days. "It's like a wobbly top," explains Simone Marchi, Lucy's deputy principal investigator. The flyby on April 20, 2025, was a dress rehearsal for Lucy's primary mission to Jupiter's **Trojan asteroids**, but it delivered a scientific jackpot. Images revealed a **bilobate** structure two lobes connected by a narrow neck, resembling a peanut. These lobes likely formed when fragments from an ancient collision gently **coalesced** under their mutual gravity. At just 155 million years old, Donaldjohanson is a cosmic toddler compared to similar asteroids Bennu and Ryugu, which are 1 to 2 billion years old. The asteroid's sluggish rotation is a recent development. Scientists estimate it spun at least 10 times faster when it formed, but a subtle force called the **YORP effect** has been applying the brakes. As sunlight warms the asteroid's **asymmetrical** surface, it radiates **infrared** heat, creating a tiny recoil that twists the rock over millions of years. This same effect can speed up rotation, as seen in Bennu (spinning once every four hours) and Ryugu (once every seven hours). Lucy's instruments also detected iron-rich clay minerals a telltale sign of liquid water in the distant past. But the water exposure was brief, unlike Bennu and Ryugu, where magnesium-rich clays indicate prolonged soaking. Marchi notes that these differences are crucial clues to our solar system's origin story. "Every subtle difference is another clue," he says. The encounter has primed the team for Lucy's next target: the Trojan asteroid Eurybates in 2027. These pristine fossils of the early solar system may rewrite what we know about planetary formation.

Word Treasure

bilobate -- Having two rounded parts connected by a narrow section, like a peanut.

YORP effect -- A force from sunlight that slowly changes how fast an asteroid spins over millions of years.

Trojan asteroids -- Groups of space rocks that travel along the same orbit as Jupiter, one group ahead and one behind the planet.

coalesced -- Came together under the influence of gravity to form a single object.

asymmetrical -- Not the same on all sides; having an uneven shape.

infrared -- A type of invisible light that we feel as heat, just beyond red in the spectrum.

Background you need

NASA's Lucy spacecraft launched in 2021 on a 12-year mission to explore Jupiter's Trojan asteroids--two swarms of space rocks that share Jupiter's orbit, one ahead and one behind the giant planet.

These asteroids are considered time capsules from the early solar system, preserving material from over 4 billion years ago.

The asteroid Donaldjohanson is named after Donald Johanson, the paleoanthropologist who discovered the famous 3.2-million-year-old 'Lucy' fossil in Ethiopia, which itself revolutionized our understanding of human evolution.

The YORP effect (named after Yarkovsky, O'Keefe, Radzievskii, and Paddack) is a subtle but powerful force that changes an asteroid's spin due to uneven heating and cooling.

NASA chose the Lucy name to honor that fossil, making the asteroid flyby a poetic connection to the mission's goal of studying ancient relics, now in space.

Break it down

LEAD: Lucy spacecraft captures wobbly peanut-shaped asteroid Donaldjohanson
  +- SPECIFIC: 'Dizzying dance' with end-over-end rotation every 10.5 days, swaying every 26.5 days
  +- QUOTE: Simone Marchi calls it 'like a wobbly top'
FLYBY CONTEXT: Dress rehearsal for Trojan mission turns into scientific jackpot
  +- WHAT: Bilobate structure -- two lobes connected by neck, formed from ancient collision coalescence
AGE & COMPARISON: Asteroid is only 155 million years old (cosmic toddler), vs. Bennu/Ryugu at 1-2 billion years
  +- WHY: Slower rotation today due to YORP effect (thermal radiation recoil over millions of years)
  +- CONTRAST: Bennu and Ryugu spun up, not down
WATER CLUES: Iron-rich clay minerals indicate brief exposure to liquid water
  +- COMPARISON: Bennu/Ryugu have magnesium-rich clays from prolonged water soaking
BIG PICTURE: Marchi says subtle differences are clues to solar system origin
  +- FUTURE: Next target Eurybates in 2027 -- pristine Trojan fossils that may rewrite planetary formation

Why it matters

Studying asteroids like Donaldjohanson helps scientists understand how our Earth formed and where water came from, and also gives us practical knowledge about deflecting potentially dangerous space rocks in the future.

Test what you remember

1. **What unusual shape did the asteroid Donaldjohanson have?**
   - Ⓐ A perfect sphere
   - Ⓑ A pancake-like disk
   - Ⓒ A two-lobed peanut shape
   - Ⓓ A cube with rounded corners
2. **How fast was the Lucy spacecraft moving during the flyby?**
   - Ⓐ 5,000 mph
   - Ⓑ 30,000 mph
   - Ⓒ 100,000 mph
   - Ⓓ 150,000 mph
3. **What force is responsible for slowing the asteroid's rotation?**
   - Ⓐ Solar wind
   - Ⓑ Magnetic braking
   - Ⓒ The YORP effect
   - Ⓓ Tidal friction
4. **What evidence did Lucy find that suggests liquid water once existed on the asteroid?**
   - Ⓐ Ancient riverbeds
   - Ⓑ Iron-rich clay minerals
   - Ⓒ Fossilized microbes
   - Ⓓ Ice caps
5. **Approximately how old is the asteroid Donaldjohanson?**
   - Ⓐ 4.5 billion years
   - Ⓑ 1.5 billion years
   - Ⓒ 155 million years
   - Ⓓ 10 million years
6. **What is Lucy's next target after this flyby?**
   - Ⓐ Bennu
   - Ⓑ Ryugu
   - Ⓒ Eurybates
   - Ⓓ Vesta

Answer key (try the quiz first!): 1: C 2: B 3: C 4: B 5: C 6: C

Questions to think about

1. **Scientific breakthrough: The Lucy flyby exceeded expectations by revealing unexpected wobbly motion and evidence of ancient water, proving that even rehearsal missions can yield major discoveries.**

2. **Limited data from fast flyby: At 30,000 mph, Lucy only had a brief window to study the asteroid, so some detailed features may have been missed; future missions might need slower orbits for thorough analysis.**

3. **Evolutionary force of the YORP effect: The YORP effect demonstrates how non-gravitational forces can dramatically shape asteroid spins over millions of years, offering a new lens to study solar system dynamics.**

4. **Path to Trojan asteroids: Lucy's next target, Eurybates in 2027, will be an untouched Trojan asteroid that could hold clues to the earliest days of the planets, and this rehearsal increases confidence in the main mission.**

MY THOUGHTS (at least 20 words):